ASSOCIATIVITY OF THE CAP ACTION WITH THE COMPOSITION OF FAMILIES OF CYCLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The setup situation for delta-dimensional cycles and Gysin maps (Chow Homology, Situation chow-situation-setup, canonical tag 02QL). cited at statement line 2923. Fixes the pair (S, delta) in which all cycle groups live.
- Locality of the cap action on open subschemes (relative-cycles chapter, Lemma lemma-action-opens, canonical tag 0H67). cited at proof lines 2935-2936. Lets the proof assume the base Z affine.
- Bilinearity of the cap action in the second variable (relative-cycles chapter, Lemma lemma-action-bilinear, canonical tag 0H66). cited at proof lines 2938-2939. Reduces the test cycle gamma to a single prime cycle.

Source reference. Original reference 02QL: sources/stacks-chow/tags.txt, source lines 3592–3592. Cited at statement line 2923: fixes the ambient setup (S, delta).

Source reference. Original lemma 0H67: sources/stacks-chow/tags.txt, source lines 21412–21412. Cited at proof lines 2935-2936: lets the proof assume Z affine.

Source reference. Original lemma 0H66: sources/stacks-chow/tags.txt, source lines 21411–21411. Cited at proof lines 2938-2939: reduces gamma to a single prime cycle.

Lemma 0.2. *Let (S, δ) be as in Chow Homology, Situation 02QL. Let $X \to Y \to Z$ be morphisms of schemes locally of finite type over S. Let $r, s, e \geq 0$. Then*

$$(\alpha \circ \beta) \cap \gamma = \alpha \cap (\beta \cap \gamma) \quad \text{in} \quad Z_{r+s+e}(X)$$

where α is a family of r-cycles on fibres of X/Y, β is a family of s-cycles on fibres of Y/Z, and $\gamma \in Z_e(Z)$.

Proof. Since we are proving an equality of cycles on X, we may work locally on Z, see Lemma 0H67. Thus we may assume Z is affine. In particular γ is a finite linear combination of prime cycles. Since $- \cap -$ is linear in the second variable (Lemma 0H66), it suffices to prove the equality when $\gamma = [W]$ for some integral closed subscheme $W \subset Z$ of δ-dimension e.

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Let $z \in W$ be the generic point. Write $\beta_z = \sum m_j[V_j]$ in $Z_s(Y_z)$. Then $\beta \cap \gamma$ is equal to $\sum m_j[\overline{V}_j]$ where $\overline{V}_j \subset Y$ is an integral closed subscheme mapped by $Y \to Z$ into W with generic fibre V_j. Let $y_j \in V_j$ be the generic point. We may and do view also as the generic point of $\overline{V}_j$ (mapping to z in W). Write $\alpha_{y_j} = \sum n_{jk}[W_{jk}]$ in $Z_r(X_{y_j})$. Then $\alpha \cap (\beta \cap \gamma)$ is equal to

$$\sum m_j n_{jk}[\overline{W}_{jk}]$$

where $\overline{W}_{jk} \subset X$ is an integral closed subscheme mapped by $X \to Y$ into $\overline{V}_j$ with generic fibre W_{jk}.

On the other hand, let us consider

$$(\alpha \circ \beta)_z = (Y_z \to Y)^*\alpha \cap \beta_z = (Y_z \to Y)^*\alpha \cap (\sum m_j[V_j])$$

By the construction of $- \cap -$ this is equal to the cycle

$$\sum m_j n_{jk}[(\overline{W}_{jk})_z]$$

on X_z. Thus by definition we obtain

$$(\alpha \circ \beta) \cap [W] = \sum m_j n_{jk}[\widetilde{W}_{jk}]$$

where $\widetilde{W}_{jk} \subset X$ is an integral closed subscheme which is mapped by $X \to Z$ into W with generic fibre $(\overline{W}_{jk})_z$. Clearly, we must have $\widetilde{W}_{jk} = \overline{W}_{jk}$ and the proof is complete. $\square$